

During my time at Iowa State, I have taken numerous classes that I would describe as helpful in developing my skills to become a professional engineer. If I was asked what classes were the most beneficial, I would not have started by listing general education courses. However, when I take a step back and think about the skills I have learned in these classes, I realize that they may be more important than I once believed. One general education class that has helped me develop as an engineer was Intro to International Studies (INST 235). This class gave me a deeper understanding of different people, cultures, and perspectives around the world. As an engineer, it can be easy to focus mostly on whether a solution technically works, but that does not necessarily mean that it will work well for every user. Different cultures and groups of people can have different needs, expectations, and ways that they interact with technology. International Studies helped me become more aware of these differences and think about who I am actually designing a solution for. Instead of assuming that every user will interact with a system in the same way that I would, I have learned to consider different perspectives and backgrounds. This can impact things such as how easy a product is to use, how information is presented, what features users actually need, and whether the solution is accessible to the people it is intended for. It also helped me understand that engineering decisions can have effects beyond the technical solution, especially when a product or system may be used by people around the world. Considering cultural and societal differences can ultimately help engineers create solutions that are more useful and effective for a wider range of people.

Another general education class that has helped me was ENGL 3140, Technical Communication. This class helped me understand how important communication is within engineering. An engineer can have a great technical solution, but it is much less useful if they cannot clearly explain the design to other people. In this class, I learned how to organize technical information and write documents in a way that is clear and easy for the intended

audience to understand. This has especially helped me when writing design documents because I have learned how to explain the purpose of a system, its requirements, design decisions, and how the different parts of the system work together. Technical Communication also helped me understand that the amount of technical detail I include should depend on who I am communicating with. I might explain something differently to another engineer than I would to a customer, manager, or someone on a different team. This is important on engineering projects where multiple teams may be working on different parts of the same system. Clear documentation and communication help everyone understand what is being built, why certain decisions were made, and what their responsibilities are. This reduces confusion between teams and makes it easier for people to work together toward the same engineering solution.

Together, these classes have helped me see that engineering involves more than just finding a technically correct solution. International Studies has helped me think more about the people and cultures that a solution will affect, while Technical Communication has helped me communicate those solutions effectively to the people involved in designing, using, and understanding them. Both of these are important because a successful engineering solution needs to not only work technically, but also meet the needs of its users and be communicated clearly to everyone involved.